

20.1 Magnetic Fields

Question Paper

Course	CIE A Level Physics
Section	20. Magnetic Fields
Topic	20.1 Magnetic Fields
Difficulty	Easy

Time allowed: 60

Score: /45

Percentage: /100

Question 1a

A long straight wire passes through a piece of card. There is a current in the wire, as shown in Fig. 1.1a.

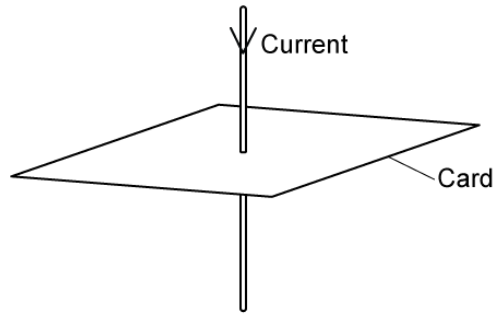


Fig. 1.1a

The current in the wire produces a magnetic field around the wire.

(i)

Fig. 1.1b shows the view of the card from above. On Fig. 1.1b, draw the magnetic field around the wire.

Show the direction of the magnetic field by drawing an arrow on each field line.

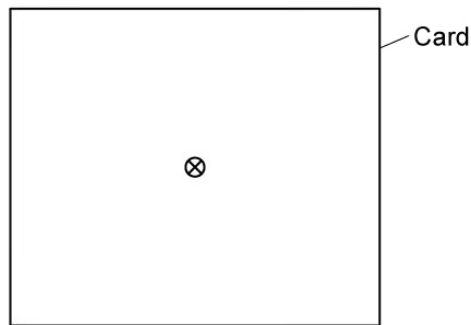


Fig. 1.1b

[2]

(ii)

Two identical wires with the same current pass through a different piece of card.

On Fig. 1.1c, draw the magnetic field around the wires.

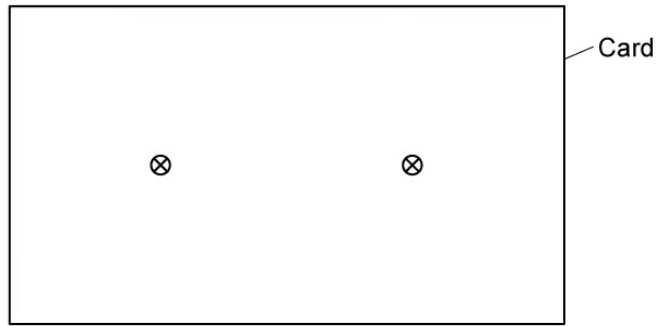


Fig. 1.1c

[2]

[4 marks]

Question 1b

The current-carrying wire is placed in a uniform magnetic field.

(i)

State the equation, and define all the variables, for the force on a current-carrying wire in a uniform magnetic field at different angles.

[2]

(ii)

Describe how the wire should be placed to experience

1. the maximum force due to the magnetic field
2. no force due to the magnetic field.

[2]

[4 marks]

Question 1c

A current-carrying conductor placed at right angles to a uniform magnetic field which has a magnetic flux density of 15×10^{-3} T.

The conductor has a length of 1.2 m and a current of 1.5 A flowing through it.

Calculate the magnetic force acting on the conductor.

[2 marks]

Question 1d

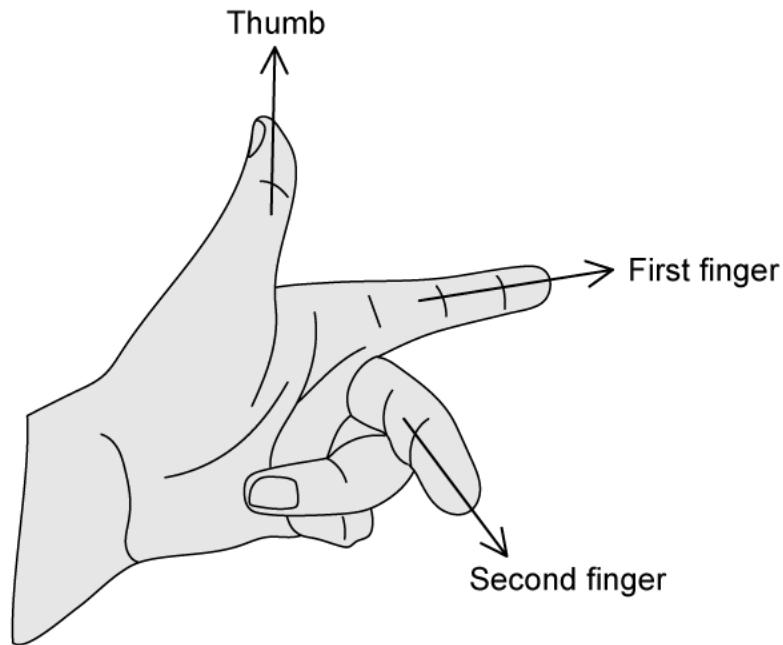
The conductor is then rotated so that it is at an angle of 30° to the magnetic field.

Calculate the new force acting on the conductor.

[2 marks]

Question 2a

When the force, the magnetic field and the current are all mutually perpendicular to each other, the directions of each can be represented by Fleming's left-hand rule, as shown in Fig. 1.1.

**Fig. 1.1**

State what is represented by the direction of

(i)
the thumb

[1]

(ii)
the first finger

[1]

(iii)
the second finger

[1]

[3 marks]

Question 2b

Fig. 1.2 shows a magnetic field.

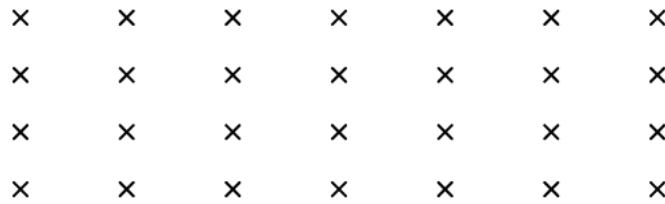


Fig. 1.2

(i)

State whether the magnetic field is acting into or out of the page.

[1]

(ii)

An electron enters the magnetic field at point A, as shown in Fig. 1.3.

On Fig. 1.3, draw

1. an arrow on the electron to show the direction of the force
2. the path of the electron through the magnetic field

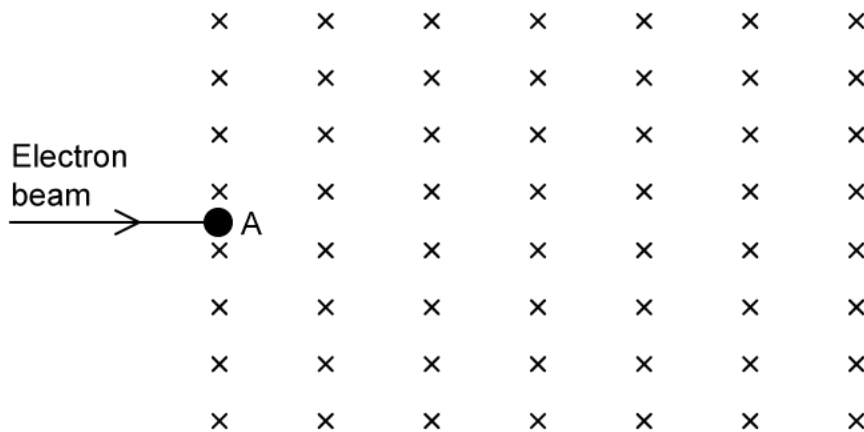


Fig. 1.3

[4]

[5 marks]

Question 2c

When a moving charge enters a magnetic field the magnetic field produces a force on the charge, which can be calculated using

$$F = Bqv$$

(i)

State the meaning of the quantities B , q and v .

[3]

(ii)

When in the field, the charge begins to move in a circular orbit.

State the name and equation of the force that causes the charge to move in a circular orbit.

[3]

[6 marks]

Question 2d

When the electron enters the magnetic field, its speed is $3.0 \times 10^6 \text{ m s}^{-1}$. The magnetic field has a magnetic flux density of $3.2 \times 10^{-3} \text{ T}$.

(i)

Using the equations in (c), show that the radius of the circular orbit of the charged object inside the magnetic field is equal to

$$r = \frac{mv}{Bq}$$

[3]

(ii)

Calculate the radius of the circular orbit of the electron

Mass of an electron = $9.11 \times 10^{-31} \text{ kg}$

Charge of an electron = $1.60 \times 10^{-19} \text{ C}$

[2]

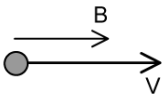
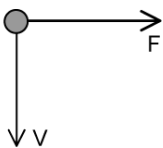
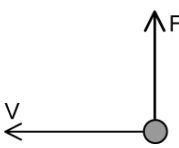
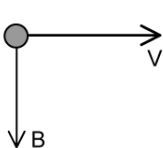
[5 marks]

Question 3a

Positive charges are seen passing through different magnetic fields.

In Table 1.1, state the direction of the missing quantity between B , v and F in each scenario

Table 1.1

		Direction
(i)		
(ii)		
(iii)		
(iv)		
(v)		

[5 marks]

Question 3b

Fig. 1.1 shows a rectangular wire loop ABCD with width 6.0 cm and length 9.0 cm in a uniform magnetic field of strength 0.75 T.

The current in the loop is 3.5 A.

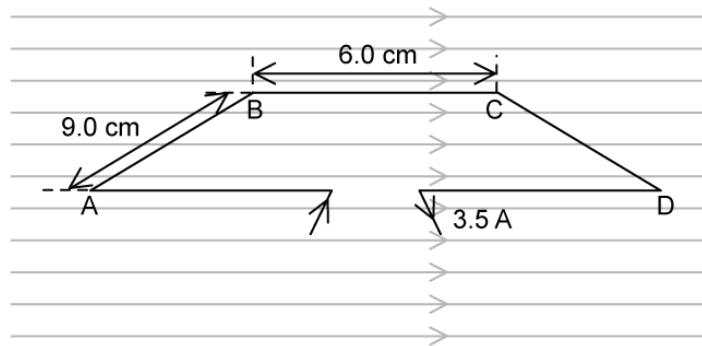


Fig. 1.1

Draw arrows on Fig. 1.1 to show the direction of the force on

(i)
side AB

[1]

(ii)
side CD

[1]

[2 marks]

Question 3c

Calculate the magnitude of the force acting on

(i)

side AB

[2]

(ii)

side BC

[1]

[3 marks]

Question 3d

(i)

When the current is switched off the loop does not move.

Describe the motion of the loop once the current is switched on.

[1]

(ii)

Determine the effect on the magnitude of the magnetic force on side CD if

1. the magnetic flux density is halved
2. the length of the wire is increased by a factor of 4
3. the current is reduced by 20%

[3]

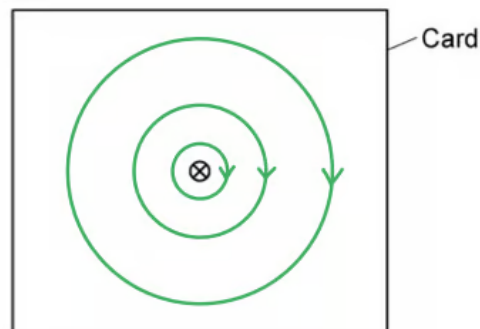
[4 marks]

Model Answers: Easy

1a

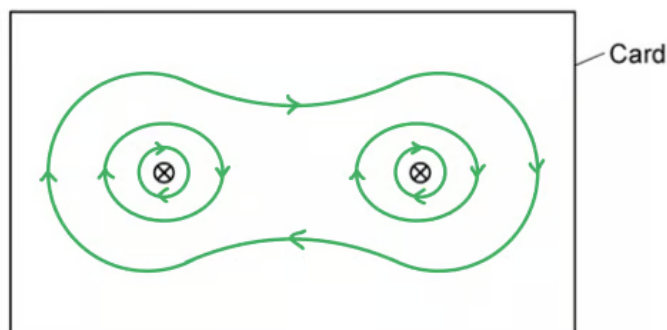
(a)

(i) The magnetic field around the wire is as follows:



- At least three circular lines drawn with increasing separation; [1 mark]
- Direction of field shown as clockwise; [1 mark]

(ii) The magnetic field around the wires is as follows:



- Clockwise magnetic field lines around the wires; [1 mark]
- At least 3 circular / elliptical lines drawn around both strips; [1 mark]

[Total: 4 marks]

The direction of the magnetic field lines around a wire is determined by the **right-**

hand grip rule

- A cross $\otimes$ means the wire is travelling **into** the page – like the back of an arrow going away from you
- A dot $\odot$ means the wire is travelling **out** of the page – like the tip of an arrow coming towards you

1b

(b)

(i) The magnetic force on a current-carrying wire is:

- $F = BIL \sin \theta$ [1 mark]
- Where B = magnetic flux density, I = current in the conductor, L = length of the conductor and θ = the angle between the wire and the field; [1 mark]

(ii) The angle θ between the wire and the magnetic field for...

1. The maximum force is when:

- $\theta = 90^\circ$ / the wire and field are perpendicular / $\sin \theta = 1$; [1 mark]

2. Zero force is when:

- $\theta = 0^\circ$ / the wire and field are parallel / $\sin \theta = 0$; [1 mark]

[Total: 4 marks]

1c

(c) Calculate the magnetic force acting on the conductor:

List the known quantities:

- Length of conductor, $L = 1.2 \text{ m}$
- Current flowing in conductor, $I = 1.5 \text{ A}$
- Magnetic flux density, $B = 15 \times 10^{-3} \text{ T}$
- Angle between conductor and field, $\theta = 90^\circ$

Use the equation for magnetic force on a current-carrying conductor:

- Magnetic force: $F = BIL \sin \theta \Rightarrow F = BIL (\sin 90^\circ = 1)$
- Magnetic force: $F = (15 \times 10^{-3}) \times 1.5 \times 1.2$ [1 mark]
- Magnetic force: $F = 0.027 \text{ N}$ [1 mark]

[Total: 2 marks]

1d

(d) Calculate the new force acting on the conductor:

List the known quantities:

- Length of conductor, $L = 1.2 \text{ m}$
- Current flowing in conductor, $I = 1.5 \text{ A}$
- Magnetic flux density, $B = 15 \times 10^{-3} \text{ T}$
- Angle between conductor and field, $\theta = 30^\circ$

Calculate the new force acting on the conductor:

- Magnetic force: $F = BIL \sin \theta$
- Magnetic force: $F = (15 \times 10^{-3}) \times 1.5 \times 1.2 \times \sin 30^\circ$ [1 mark]
- Magnetic force: $F = 0.0135 = 0.014 \text{ N}$ [1 mark]

[Total: 2 marks]

2a

(a)

(i) The thumb represents the:

- Motion / Force; [1 mark]

(ii) The first finger represents the:

- Magnetic field; [1 mark]

(iii) The second finger represents the:

- Current; [1 mark]

[Total: 3 marks]

It is important not to get mixed up with what each finger represents. An easy way to remember this is:

- Thu**M**b = **M**otion / force
- **F**irst finger = (magnetic) **F**ield
- Se**C**ond finger = **C**urrent

2b

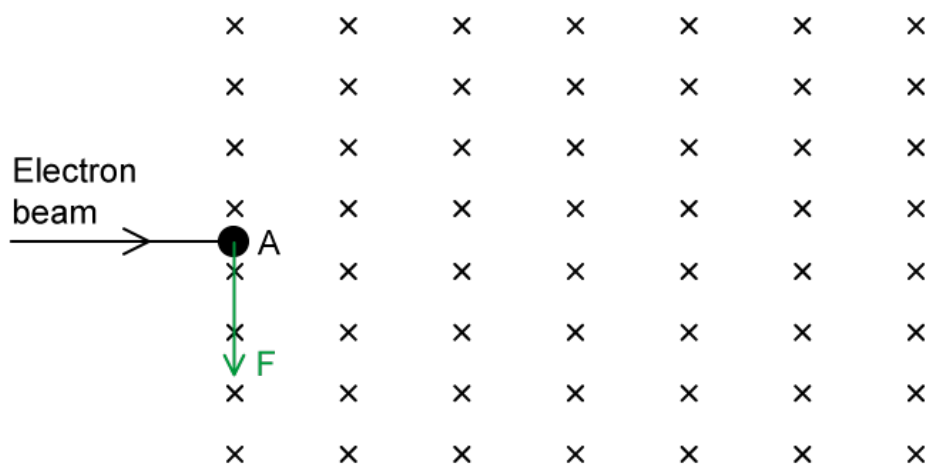
(b)

(i) The magnetic field is acting:

- Into the page; [1 mark]

(ii) The force and motion of the electron are as follows:

1. The force on the electron

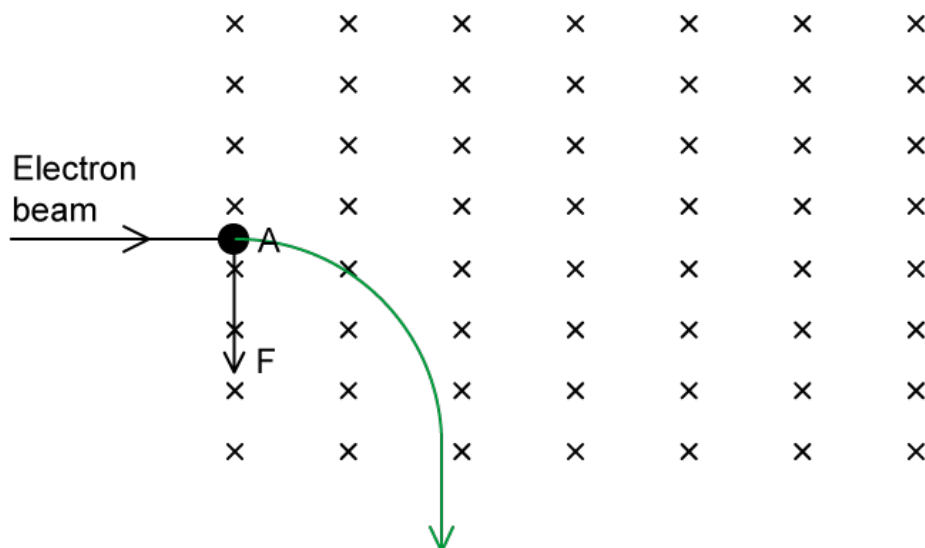


- Arrow perpendicular to electron beam; [1 mark]
- Pointing downwards; [1 mark]

2. The motion of the electron

- Arrow perpendicular to electron beam; [1 mark]
- Pointing downwards; [1 mark]

2. The motion of the electron



- Moves in a circular arc; [1 mark]
- Leaves the field pointing downwards; [1 mark]

[Total: 5 marks]

We can use Fleming's left-hand rule to determine the direction of the force acting on the electron, however, as we have an isolated charge the second finger gives the direction of the **velocity** of a **positive** charge.

As we have a **negatively-charged** electron we reverse the direction of the velocity

- The second finger (motion) will point to the left (since the electron, or negative charge, is moving to the right)
- The first finger (field) points into the page
- Hence the thumb (force) will point downwards

When the electron is in the magnetic field the force acting on it will be perpendicular to the direction in which it is moving. This causes the electron to move in a **circular path**.

The force exerted on the electron from the magnetic field will point towards the centre of the circular path, hence in this example, the circular arc is directed downwards

2c

(c)

(i) The symbols represent the following quantities:

- B = magnetic flux density of the field; [1 mark]
- q = charge of the particle; [1 mark]
- v = velocity of the particle; [1 mark]

(ii) The force that causes the charge to move in a circular orbit is the:

- Centripetal force; [1 mark]

The equation to calculate the centripetal force is:

- $F = \frac{mv^2}{r}$ [1 mark]
- Where m = mass of the charge, v = velocity, and r = radius of the orbit; [1 mark]

[Total: 6 marks]

2d

(d)

(i) Use the equations from (c) to derive the equation for radius:

- Magnetic force on a charge: $F = Bqv$
- Centripetal force: $F = \frac{mv^2}{r}$

Equate the two expressions of force:

- Equate: $Bqv = \frac{mv^2}{r}$ [1 mark]

- Rearrange: $r = \frac{mv^2}{Bqv}$ [1 mark]
- Simplify: $r = \frac{mv}{Bq}$ [1 mark]

(ii) Calculate the radius of the circular orbit of the electron:

List the known quantities:

- Mass of an electron, $m = 9.11 \times 10^{-31} \text{ kg}$
- Charge of an electron, $q = 1.60 \times 10^{-19} \text{ C}$
- Magnetic flux density, $B = 3.2 \times 10^{-3} \text{ T}$
- Speed of the electron, $v = 3.0 \times 10^6 \text{ m s}^{-1}$

Use the equation from part (i):

- $r = \frac{mv}{Bq}$
- $r = \frac{(9.11 \times 10^{-31}) \times (3.0 \times 10^6)}{(3.2 \times 10^{-3}) \times (1.60 \times 10^{-19})}$ [1 mark]
- $r = 5.3 \times 10^{-3} \text{ m}$ [1 mark]

[Total: 5 marks]

3a

(a)

(i) When the field and motion are parallel:

- Force = 0 / no direction; [1 mark]

(ii) When the field and motion are opposed:

- Force = 0 / no direction; [1 mark]

(iii) When the force is to the left and the motion is down:

- Magnetic field / B is into the page; [1 mark]

(iv) When the force is up and the motion is to the right:

- Magnetic field / B is out of the page; [1 mark]

(v) When the motion is to the left and the field is down:

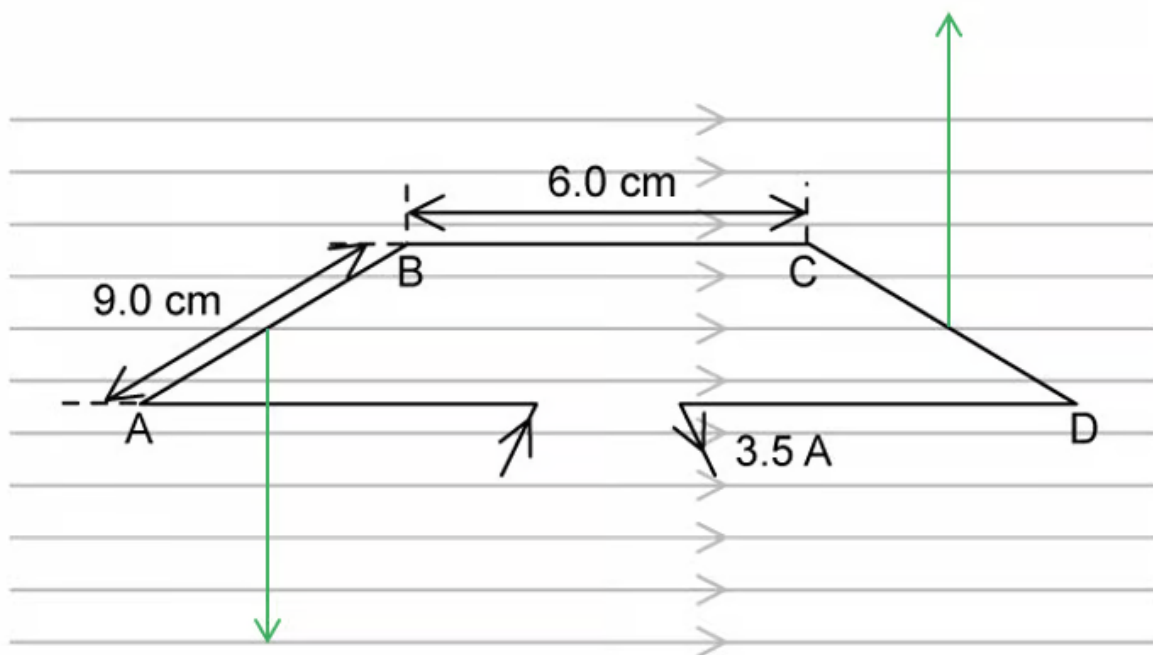
- Force / F is into the page; [1 mark]

[Total: 5 marks]

Remember - there is only a force on a charge in a magnetic field if the current is at **right angles** to the magnetic field.

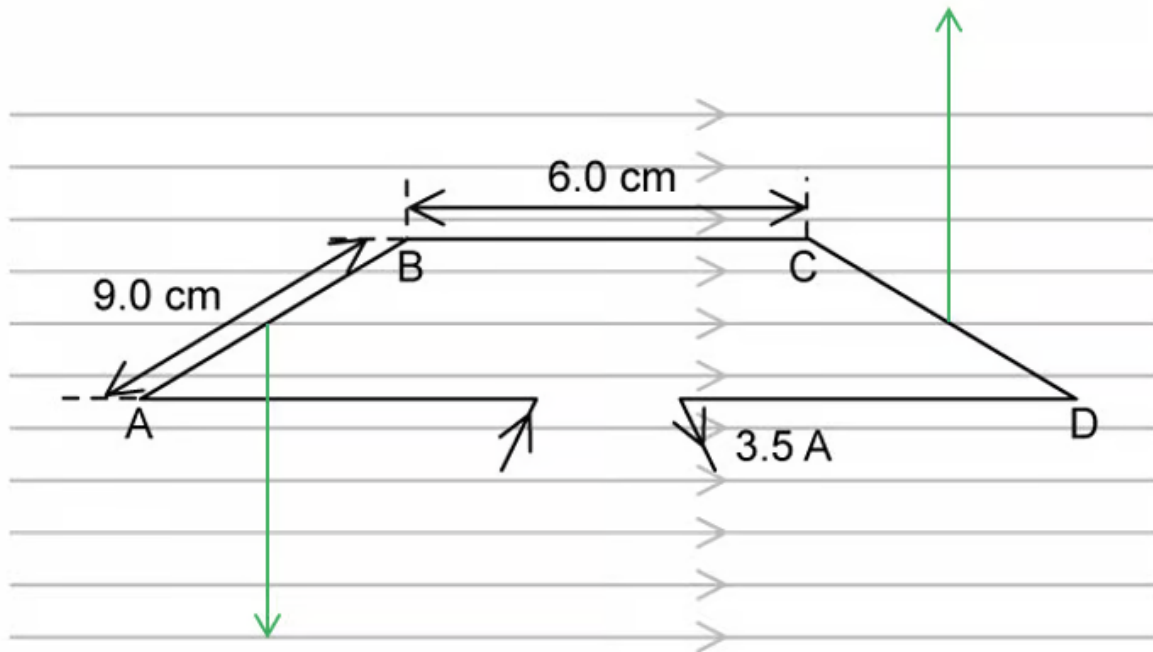
3b

(b) The direction of the force on each side is...



(i) Side AB

- Arrow pointing down; [1 mark]



(i) Side AB

- Arrow pointing down; [1 mark]

(ii) Side CD

- Arrow pointing up; [1 mark]

[Total: 2 marks]

3c

(c)

(i) Calculate the force on side AB:

List the known quantities:

- Length, $L = 9.0 \text{ cm} = 0.09$
- Magnetic field strength, $B = 0.75 \text{ T}$
- Current, $I = 3.5 \text{ A}$
- Angle between wire and field, $\theta = 90^\circ$

Use the equation for force on a current-carrying conductor:

- Magnetic force: $F = BIL \sin \theta \Rightarrow F = BIL (\sin 90^\circ = 1)$
- $F = 0.75 \times 3.5 \times 0.09$ [1 mark]
- $F = 0.236 = 0.24 \text{ N}$ [1 mark]

(ii) Calculate the force on side BC:

- Magnetic force: $F = BIL \sin \theta \Rightarrow F = 0 (\sin 0^\circ = 0)$ [1 mark]

[Total: 3 marks]

3d

(d)

(i) When the current is switched on:

- The loop rotates in an anti-clockwise direction; [1 mark]

(ii) The effect on the magnetic force when...

1. the magnetic flux density is halved

- $F = BIL \Rightarrow F \propto B$
- (Hence) the force will be halved
- $F = \frac{1}{2} \times 0.16 = 0.08 \text{ N}$ [1 mark]

2. the length of the wire is increased by a factor of 4

- $F = BIL \Rightarrow F \propto L$
- (Hence) the force will increase by a factor of 4